

196. If $2x^2 + 12x + 18 = 0$, what is the value of x ?

(Bank P.O., 2006)

- (a) -3 (b) -2
(c) 2 (d) 3
(e) More than one answer

197. If $x^2 - 7x = -12$, what is the value of x ? (C.E.T., 2006)

- (a) -3 or -4 (b) 3 or 4
(c) 3 or -4 (d) Cannot be determined
(e) None of these

198. The roots of the equation $2x^2 - 11x + 15 = 0$ are

(Campus Recruitment, 2008)

- (a) $3, \frac{5}{2}$ (b) $5, \frac{3}{2}$
(c) $-3, -\frac{5}{2}$ (d) None of these

199. The value of $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) \cdots \left(1 - \frac{1}{m}\right)$ is

(P.C.S., 2008)

- (a) 1 (b) $\frac{1}{m}$
(c) $\frac{1}{2m}$ (d) $\frac{1}{1 \cdot 2 \cdot 3 \cdots (m-1)m}$

200. The expression $\left(1 + \frac{1}{3}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{5}\right) \cdots \left(1 + \frac{1}{n}\right)$

simplifies to

(M.B.A. 2006, 2008)

- (a) $\frac{n+1}{3}$ (b) $\frac{n}{n+1}$
(c) $\frac{3}{n}$ (d) $1 + \frac{1}{3} \cdot \frac{1}{4} \cdot \frac{1}{5} \cdots \frac{1}{n}$

201. When simplified, the product $\left(2 - \frac{1}{3}\right)\left(2 - \frac{3}{5}\right)$

$\left(2 - \frac{5}{7}\right) \cdots \left(2 - \frac{997}{999}\right)$ is equal to (R.R.B., 2007)

- (a) $\frac{5}{999}$ (b) $\frac{1001}{999}$
(c) $\frac{1}{1001}$ (d) $\frac{1001}{3}$

202. $\frac{3}{4}\left(1 + \frac{1}{3}\right)\left(1 + \frac{2}{3}\right)\left(1 - \frac{2}{5}\right)\left(1 + \frac{6}{7}\right)\left(1 - \frac{12}{13}\right) = ?$

- (a) $\frac{1}{5}$ (b) $\frac{1}{6}$
(c) $\frac{1}{7}$ (d) None of these

203. The value of $\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right)\left(1 - \frac{1}{5^2}\right) \cdots \left(1 - \frac{1}{11^2}\right)$

$\left(1 - \frac{1}{12^2}\right)$ is

(a) $\frac{11}{20}$ (b) $\frac{13}{15}$

(c) $\frac{13}{18}$ (d) $\frac{15}{16}$

(e) None of these

204. What is the value of the following expression?

$$\frac{1}{(2^2 - 1)} + \frac{1}{(4^2 - 1)} + \frac{1}{(6^2 - 1)} + \cdots + \frac{1}{(20^2 - 1)}$$

(a) $\frac{9}{19}$ (b) $\frac{10}{19}$

(c) $\frac{10}{21}$ (d) $\frac{11}{21}$

205. Find the sum : $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72}$

$$+ \frac{1}{90} + \frac{1}{110} + \frac{1}{132}.$$

(a) $\frac{7}{8}$ (b) $\frac{11}{12}$

(c) $\frac{15}{16}$ (d) $\frac{17}{18}$

206. The sum of the first 35 terms of the series

$$\frac{1}{2} + \frac{1}{3} - \frac{1}{4} - \frac{1}{2} - \frac{1}{3} + \frac{1}{4} + \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \cdots$$

(a) $-\frac{1}{2}$ (b) $-\frac{1}{4}$

(c) $\frac{1}{4}$ (d) None of these

207. $\left(1\frac{1}{2} + 11\frac{1}{2} + 111\frac{1}{2} + 1111\frac{1}{2}\right)$ is equal to (S.S.C., 2010)

(a) 617 (b) 618

(c) $1234\frac{1}{2}$ (d) 1236

208. $\left(999\frac{999}{1000} \times 7\right)$ is equal to (C.P.O., 2007)

(a) $6633\frac{7}{1000}$ (b) $6993\frac{7}{1000}$

(c) $6999\frac{993}{1000}$ (d) $7000\frac{7}{1000}$

209. The value of $999\frac{995}{999} \times 999$ is

(a) 990809 (b) 998996

(c) 998999 (d) 999824

210. $\left(999\frac{1}{7} + 999\frac{2}{7} + 999\frac{3}{7} + 999\frac{4}{7} + 999\frac{5}{7} + 999\frac{6}{7}\right)$

is simplified to

(a) 2997 (b) 5979

(c) 5994 (d) 5997